

Mohammed Abdul Omer

AI and ML Engineer

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LinkedIn: linkedin.com/in/mohammad-abdul-omer GitHub: github.com/MOHD-OMER

Portfolio: mohdomer.vercel.app Hugging Face: huggingface.co/mohdomer

PROFESSIONAL SUMMARY

Computer Science student (graduating May 2026, AI and ML specialization) with hands-on experience designing, fine-tuning, and shipping production-grade Generative AI systems, Retrieval-Augmented Generation (RAG) pipelines, and multi-agent LLM workflows. Proficient across the full machine learning (ML) lifecycle from model training and evaluation to MLOps, containerized API deployment, and live drift monitoring. Strong communicator and collaborative problem-solver committed to building scalable, reliable AI solutions.

TECHNICAL SKILLS

Languages: Python, SQL

ML and Deep Learning: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, OpenCV, YOLO (object detection), MobileNet (image classification), CNNs, Grad-CAM

LLM and Generative AI: LangChain, LangGraph, CrewAI, QLoRA, PEFT, Mistral 7B, LLaMA 3, Groq, Ollama, RAG, ChromaDB, BM25, Prompt Engineering

Backend and MLOps: FastAPI, Django, Streamlit, Gradio, MLflow, Weights and Biases, Evidently AI, Docker

Tools: Git, GitHub, Hugging Face Hub, Kaggle, Vercel

Soft Skills: Cross-functional collaboration, technical documentation, problem solving, project ownership, rapid prototyping, adaptability

PROFESSIONAL EXPERIENCE

AI Engineer Intern | TechZone Software Academy | Hyderabad, India | March 2025 – June 2025

- Architected and delivered 4 end-to-end production AI systems for a live educational platform serving over 300 students and educators, reducing manual academic workload by an estimated 40 percent across 3 departments
- Engineered a Retrieval-Augmented Generation (RAG) tutoring assistant using LangChain, ChromaDB vector database, and LLaMA 3 via Groq API, cutting average educator query response overhead and improving student support coverage across subjects
- Implemented an automated language test engine with LLM-based scoring across reading, writing, and listening modules, and a question generation plus attendance automation system; orchestrated all 4 services as production-ready FastAPI REST APIs

AI and ML Intern | RAM Innovative Infotech | Hyderabad, India | November 2024

- Developed and evaluated a supervised disease-prediction classifier using Scikit-learn on a structured medical dataset, then integrated the trained model into a live Django web application serving real users

PROJECTS

Multi-Document RAG Chatbot | *LangChain, LangGraph, ChromaDB, BM25, FlashRank, FastAPI, Streamlit*

- Engineered a hybrid retrieval agent combining dense vector embeddings (semantic search) with BM25 sparse keyword search and FlashRank cross-encoder reranking; built an async FastAPI backend with server-sent event (SSE) streaming that reduced response latency on multi-document queries; deployed on Hugging Face Spaces

LLM Fine-Tuning: Mistral 7B Medical QA | *QLoRA, PEFT, SFTTrainer, Weights and Biases*

huggingface.co/mohdomer/mistral-7b-medical-qa-qlora

- Fine-tuned Mistral 7B Instruct using QLoRA (4-bit NF4 quantization) and SFTTrainer on a Kaggle T4 GPU; tracked all experiments in Weights and Biases and achieved consistent ROUGE and BLEU score improvements over the untuned baseline; published model to Hugging Face Hub

Multi-Agent Research System | *CrewAI, Tavily, LLaMA 3, Python*

huggingface.co/spaces/mohdomer/nexus-research

- Orchestrated a CrewAI multi-agent pipeline with live Tavily web search, coordinating specialized agents for planning, retrieval, and synthesis to autonomously generate structured research reports

AI Orchestrator — Multi-Provider LLM Router | *FastAPI, Groq, Gemini, OpenRouter, Ollama*

ai-orchestrator-nu86.onrender.com

- Designed a unified LLM routing system supporting 17+ models across 4 providers with automatic task classification

and a benchmark mode for side-by-side latency and quality comparison

MLOps Pipeline | *MLflow, Weights and Biases, Evidently AI, Docker*

- Built a modular, reproducible ML pipeline integrating MLflow experiment tracking, model registry, and Evidently AI automated data drift monitoring; containerized the full workflow with Docker for consistent, repeatable deployments

PulmoScan AI — TB Detection System | *TBNet CNN, Grad-CAM, FastAPI*

pulmoscanai-app.hf.space

- Trained a tuberculosis detection CNN with Grad-CAM saliency maps for clinical explainability and automated PDF diagnostic report generation; deployed as a live Hugging Face Space application

TruthLens — Fake News Detector | *CNN-LSTM, NLP, Keras*

truthlens-uqo4.onrender.com

- Trained a CNN-LSTM text classifier on 40,000+ labeled news articles achieving 94.2% accuracy; deployed as a live inference web application on Render

Building Safety Smoke Detection | *YOLOv8, MobileNetV2, Django* | *B.E. Major Project*

- Trained 7 ML classifiers achieving AUC-ROC above 0.999 on 62,630 IoT sensor readings; deployed MobileNetV2 CNN reaching 96.98% validation accuracy with YOLOv8 real-time fire and smoke detection in a role-based Django application on Railway

EDUCATION

Bachelor of Engineering, Computer Science Engineering (AI and ML)

Expected May 2026

Lords Institute of Engineering and Technology, Hyderabad, India

Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Data Structures and Algorithms, Database Management Systems

ACHIEVEMENTS

- Published fine-tuned Mistral 7B medical QA model to Hugging Face Hub; model tracked with Weights and Biases showing consistent ROUGE and BLEU improvements
- Deployed 6 live AI applications across Hugging Face Spaces, Railway, and Render serving real users in production
- Completed Machine Learning Specialization (DeepLearning.AI / Andrew Ng, 2024), Generative AI with LLMs (DeepLearning.AI, 2024), and Python for Everybody (University of Michigan, 2023)